

For our notation, let X be the number of sixes obtained in the 3 rolls. Then X has a binomial distribution with $n = 3$ and $p = \frac{1}{6}$. Hence

$$P(X = 2) = p(2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 0.06944$$

Poisson Distribution

The PMF of the Poisson distribution is given by:

$$p(x) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, 2, \dots$$

A random variable that has pmf of the form $p(x)$ is said to have **Poisson distribution** with parameter λ and any such $p(x)$ is called a **Poisson pmf** with parameter λ

Example

Let X be the number of automobile accidents at a busy intersection per week. Suppose that X has a Poisson distribution with $\lambda = 2$. Then the expected number of accidents per week is 2 and the standard deviation is $\sqrt{2}$. Calculate the probability of at least one accident in a week.

Solution

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-2} = 0.8647$$

Basic Concepts of Hypothesis Testing

Hypothesis testing is a systematic statistical procedure used to determine whether a claim about a population parameter is supported by sample evidence.

Core Terminology

- **Population Parameter:** A numerical characteristic of a population (e.g., population mean μ , population proportion p).
- **Sample Statistic:** A numerical characteristic calculated from a sample (e.g., sample mean $\bar{x}$, sample proportion $\hat{p}$).
- **Level of Significance (α):** The probability of rejecting a true null hypothesis. Commonly set at 0.05 (5%) or 0.01 (1%).
- **P-value:** The probability of obtaining a test statistic at least as extreme as the observed one, assuming the null hypothesis is true.
- **Critical Value:** A boundary value that separates the rejection region from the non-rejection region.

The Hypotheses: Null vs. Alternative

- **The Null Hypothesis (H_0):** Represents a statement of **no effect, no difference**, or status quo. It always contains an equality sign ($=$, $\leq$, or $\geq$). It is assumed true until proven otherwise.
- **The Alternative Hypothesis (H_1 or H_a):** Represents the claim the researcher wants to prove (**effect exists**). It contains inequalities ($<$, $>$, or $\neq$).

Type I and Type II Errors

Because hypothesis testing relies on sample data rather than complete census information, errors in decision-making are possible.

	Fail to Reject H_0	Reject H_0
H_0 is True	Correct Decision ($1 - \alpha$)	Type I Error (α)
H_0 is False	Type II Error (β)	Correct Decision ($1 - \beta$) (Power)

Definitions and Implications

- **Type I Error (α):** Occurs when you **reject the null hypothesis when it is actually true** (a False Positive).
 - *Probability:* $P(\text{Type I Error}) = \alpha$ (The significance level).
 - *Example:* Convicting an innocent person in court.

- **Type II Error (β):** Occurs when you **fail to reject the null hypothesis when it is actually false** (a False Negative).
 - *Probability:* $P(\text{Type II Error}) = \beta$.
 - *Example:* Acquitting a guilty person in court.
- **Power of a Test ($1 - \beta$):** The probability of correctly rejecting a false null hypothesis.

One-Tailed vs. Two-Tailed Tests

The direction of the alternative hypothesis determines the allocation of the rejection region.

- **Two-Tailed Test:** Used when checking if a parameter is simply **different** or has changed.

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0$$

The rejection region is split equally into both tails ($\alpha/2$ in each tail).

- **Left-Tailed Test:** Used when checking if a parameter is **less than** a threshold.

$$H_0 : \mu \geq \mu_0 \quad \text{vs.} \quad H_1 : \mu < \mu_0$$

The entire rejection region (α) is in the left tail.

- **Right-Tailed Test:** Used when checking if a parameter is **greater than** a threshold.

$$H_0 : \mu \leq \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0$$

The entire rejection region (α) is in the right tail.

Tests Involving the Normal Distribution (Z-Tests)

A Z-test is applicable when the population standard deviation (σ) is known, or when the sample size is large ($n \geq 30$).

Case A: Z-Test for a Single Mean (μ)

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Case B: Z-Test for a Single Proportion (p)

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \quad \text{where } \hat{p} = \frac{x}{n}$$

Small Sample Tests: Student's t-Tests

One-Sample t-Test

Used when σ is unknown, $n < 30$, and the population is approximately normal.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \text{with } df = n - 1$$

Two-Sample Independent t-Test (Equal Variances Assumed)

Used to compare the means of two distinct, independent populations (μ_1 and μ_2).

- **Hypotheses:** $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_1 : \mu_1 - \mu_2 \neq 0$
- **Test Statistic:**

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- **Pooled Standard Deviation (s_p):**

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- **Degrees of Freedom:** $df = n_1 + n_2 - 2$

Practical Examples & Step-by-Step Solutions

Example 1: Two-Sample Independent t-Test

Problem Statement: An educator wants to compare the effectiveness of two different training methods. Group 1 ($n_1 = 10$) uses Method A, and Group 2 ($n_2 = 12$) uses Method B. The exam results are summarized below:

Group 1 (Method A): $\bar{x}_1 = 78.4$, $s_1 = 5.2$, $n_1 = 10$

Group 2 (Method B): $\bar{x}_2 = 82.1$, $s_2 = 4.8$, $n_2 = 12$

Test whether there is a significant difference between the two training methods at a 5% significance level ($\alpha = 0.05$). Assume equal variances.

Solution:

1. Formulate Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad (\mu_1 - \mu_2 = 0)$$

$$H_1 : \mu_1 \neq \mu_2 \quad (\mu_1 - \mu_2 \neq 0) \quad [\text{Two-tailed test}]$$

2. Calculate the Pooled Standard Deviation (s_p):

$$s_p = \sqrt{\frac{(10-1)(5.2)^2 + (12-1)(4.8)^2}{10+12-2}}$$

$$s_p = \sqrt{\frac{9(27.04) + 11(23.04)}{20}} = \sqrt{\frac{243.36 + 253.44}{20}} = \sqrt{\frac{496.8}{20}} = \sqrt{24.84} \approx 4.984$$

3. Determine Degrees of Freedom and Critical Value:

$$df = n_1 + n_2 - 2 = 10 + 12 - 2 = 20$$

From the Student's t -table, the critical values for a two-tailed test with $df = 20$ at $\alpha = 0.005$ are ± 2.086 .

4. Calculate the Test Statistic (t_{calc}):

$$t_{calc} = \frac{78.4 - 82.1}{4.984 \sqrt{\frac{1}{10} + \frac{1}{12}}} = \frac{-3.7}{4.984 \sqrt{0.1 + 0.0833}}$$

$$t_{calc} = \frac{-3.7}{4.984 \sqrt{0.1833}} = \frac{-3.7}{4.984(0.4282)} = \frac{-3.7}{2.134} \approx -1.734$$

- 5. Conclusion:** Since $|t_{calc}| = 1.734$ is less than the critical value 2.086, it does not fall into the rejection region. We **fail to reject** H_0 . There is insufficient statistical evidence at the 5% level to conclude that a significant performance difference exists between Method A and Method B.

Example 2: One-Tailed Z-Test for a Population Proportion

Problem Statement: A politician claims that voter support for her policy exceeds 60%. Out of a random sample of 150 voters, 102 declare support. Test her claim at a 1% level of significance.

Solution:

1. Formulate Hypotheses:

$$H_0 : p = 0.60 \quad \text{vs.} \quad H_1 : p > 0.60 \quad [\text{Right-tailed test}]$$

2. Identify Values and Sample Proportion:

$$p_0 = 0.60, \quad n = 150, \quad \alpha = 0.01 \implies \hat{p} = \frac{102}{150} = 0.68$$

3. **Critical Value:** For a right-tailed Z -test at $\alpha = 0.01$, $Z_{crit} = 2.33$.

4. **Calculate Test Statistic:**

$$Z_{calc} = \frac{0.68 - 0.60}{\sqrt{\frac{0.60(1-0.60)}{150}}} = \frac{0.08}{\sqrt{\frac{0.24}{150}}} = \frac{0.08}{0.04} = 2.00$$

5. **Conclusion:** Since $Z_{calc} = 2.00 < 2.33$, we **fail to reject** H_0 . The data does not provide sufficient evidence at the 1% level to state that support significantly exceeds 60%.